

ONLINE EXAM PORTAL

Software Requirements Specification (SRS)

1. User Requirements

The following table lists the key requirements expressed from the perspective of the system's end users.

ID	User Requirement
UR1	The system shall allow students to securely log in to the online examination portal.
UR2	The system shall allow students to view available exams along with their schedules.
UR3	The system shall allow students to attend exams online within the specified time limit.
UR4	The system shall display multiple-choice and other question types clearly to students.
UR5	The system shall automatically save a student's answers during the examination.
UR6	The system shall automatically submit the exam when the allotted time expires.
UR7	The system shall automatically evaluate objective-type questions.
UR8	The system shall allow students to view their results after evaluation is complete.
UR9	The system shall allow teachers/administrators to create, modify, and manage examinations.
UR10	The system shall generate examination reports and student performance statistics.

2. Functional Requirements

The following table lists the core functional requirements that define what the system must do.

ID	Functional Requirement
FR1	The system shall authenticate users using a registered username/email and password.
FR2	The system shall provide separate access privileges for students, teachers, and administrators.
FR3	The system shall display available exams with subject, date, duration, and number of questions.
FR4	The administrator shall be able to create an exam by specifying its title, duration, marks, and schedule.
FR5	The teacher shall be able to add, edit, and delete questions from the question bank.
FR6	The system shall provide a countdown timer showing the remaining examination time.
FR7	The system shall automatically save answers whenever a student selects or changes a response, and allow navigation between questions.
FR8	The system shall prevent access to an exam before its scheduled start time and auto-submit it when the timer reaches zero.
FR9	The system shall automatically calculate the student's score for objective questions and store responses and results in a database.
FR10	The administrator shall be able to generate reports containing student performance and examination statistics.

3. Non-Functional Requirements

The following table lists the quality attributes and constraints the system must satisfy.

ID	Category	Non-Functional Requirement
NFR1	Security	The system shall prevent unauthorized users from viewing questions or examination results.
NFR2	Security	The system shall encrypt sensitive user credentials and examination data.
NFR3	Availability	The system shall be available throughout the scheduled examination period with minimal downtime.
NFR4	Performance	The system shall respond to normal user actions within a reasonable time.
NFR5	Scalability	The system shall support multiple students taking an examination simultaneously.
NFR6	Usability	The interface shall be simple, user-friendly, and responsive on computers, tablets, and mobile devices.
NFR7	Reliability	The system shall maintain data consistency if a student temporarily loses network connectivity.
NFR8	Reliability	The system shall provide backup and recovery mechanisms for examination and result data.
NFR9	Maintainability	The system shall be designed in a modular way to allow easy updates and future enhancements.
NFR10	Auditability	The system shall maintain logs of login, exam start, submission, and other key activities.

4. Requirement Elicitation and Analysis Flow

Stakeholders

Students, Teachers/Examiners, and Administrators.

Elicitation Methods

Interviews, questionnaires, observation, surveys, and brainstorming sessions were used to gather requirements from stakeholders.

Analysis Flow

1. Elicitation: Requirements were gathered from students, teachers, and administrators using the methods above.
2. Analysis: Gathered inputs were reviewed, categorized, and checked for conflicts, redundancy, and feasibility.
3. Specification: Validated needs were documented as user, functional, and non-functional requirements.
4. Validation: Requirements were reviewed with stakeholders to confirm they accurately reflect the intended system behavior.

Resulting User Requirements

Conduct exams online, manage questions, automatically evaluate exams, view results, generate reports, and maintain security.

Resulting Functional / Non-Functional Requirements

Login authentication, question bank management, timer-based exam delivery, auto-save and auto-submit, automatic evaluation, result and report generation, access control, and data security/backup.